

Transforming CT imaging with deep learning: Noise reduction, artifact management, and clinical applications – A comprehensive review

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ARTICLE INFO

Keywords:

Deep learning reconstruction
Iterative reconstruction
Computed tomography
Noise reduction
Artifact suppression
Dose optimization

ABSTRACT

Deep Learning Reconstruction (DLR) is emerging as a significant advancement in Computed Tomography (CT) imaging by addressing crucial issues such as noise reduction, artifact suppression, and dose optimization. Traditional CT reconstruction methods, namely Filtered Back Projection (FBP) and the relatively newer Iterative Reconstruction (IR), have limitations in preserving image quality as radiation doses are lowered. While FBP allows for rapid image generation, it is highly prone to image noise, often requiring higher radiation doses to maintain diagnostic quality. In contrast, IR employs iterative refinement techniques to reduce noise and enhance image quality, but this can lead to unnatural textures that undermine diagnostic confidence especially at lower doses. DLR has shown potential for clinical applications across various imaging subspecialties, such as neuro, thoracic, abdominopelvic, cardiovascular, and pediatric imaging. DLR improves lesion detection, increases soft-tissue Contrast-to-noise, and supports lower-dose protocols, which are particularly important for pediatric patients. Compared to IR, DLR retains more fine anatomical details while effectively reducing artifacts such as beam hardening and motion distortions. However, DLR also faces challenges related to model interpretability, dataset diversity, and computational resource requirements. Addressing these issues through adaptive learning models, explainable AI frameworks, and cross-institutional data sharing will be essential for broader adoption. As DLR continues to advance, its integration with AI-driven diagnostic tools and multi-modality imaging may contribute to advancing the future of CT imaging, enhancing both diagnostic accuracy and patient safety.

Introduction

Computed Tomography (CT) has transformed modern diagnostics by providing rapid, high-resolution imaging essential for a variety of conditions, from trauma to cancer [1,2]. The process of CT image reconstruction has evolved significantly, progressing from the traditional Filtered Back-Projection (FBP) technique to Iterative Reconstruction (IR), and now to Deep Learning Reconstruction (DLR) [3]. This progression has been fuelled by rapid technological advancements

throughout the computer and scanner industry and the constant desire to improve patient care, particularly in challenging scanning scenarios such as with lower-dose protocols.

For decades, FBP was the dominant CT reconstruction technique due to its computational efficiency and straightforward conceptual model [4,5]. While FBP can deliver accurate image reconstruction under ideal conditions—with high photon counts and minimal noise—these optimal situations are often difficult to achieve in clinical practice. Factors such as patient size, positioning and motion can compromise the quality of

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<https://doi.org/10.1016/j.ejrai.2025.100042>

Received 6 May 2025; Received in revised form 14 August 2025; Accepted 15 September 2025

Available online 18 September 2025

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images. Additionally, FBP typically requires relatively higher radiation dose to produce acceptable images, which restricts its application in settings where dose reduction is critical, such as in paediatric imaging and for patients undergoing multiple scans.

To address the limitations of FBP, iterative reconstruction (IR) was introduced [6–8]. Adaptive statistical image reconstruction (IR) methods concentrated on noise modeling to enhance image quality at low doses, but they often fell short in spatial resolution. Model-based IR improved noise suppression and edge preservation by incorporating system physics, yet this came with increased computational demands and resulted in an "unnatural" texture—often described as "blotchy" or "plastic-like" [9]. Hybrid IR techniques were developed to address these issues by merging statistical noise reduction with aspects of model-based reconstruction, seeking to maintain detail while reducing artificial textures [6,7,10]. Nevertheless, the challenge persists, particularly in low-dose situations, where artificial textures may still impact diagnostic accuracy and dose optimization efforts [11–13].

The limitations of FBP and IR led to the development of a novel methodology designed to overcome these issues. DLR has emerged as an innovative alternative, utilizing neural networks trained on high-quality CT images to improve the quality of low-dose scans [14]. This review assesses the technical principles of DLR, its applications across various clinical domains, and its comparative advantages over FBP and IR. We analyze its performance in neuroimaging, abdominal, and cardiovascular CT, highlighting its potential for low-dose protocols that are critical for patient safety. Additionally, we address practical considerations, limitations, and future developments, emphasizing the DLR transformative role of DLR in CT imaging and its promise for broader clinical integration.

Evolution of CT reconstruction techniques

Since its introduction in the 1970s, CT has made significant advancements in reconstruction techniques aimed at balancing image quality with radiation safety [15,16]. Achieving this balance is essential for minimizing radiation exposure, particularly for sensitive populations, yet it remains a challenge to do so without sacrificing diagnostic detail. Over the decades, CT reconstruction algorithms have progressed from FBP to IR, and most recently, to DLR. Each phase of this evolution marks a pivotal advancement, addressing the shortcomings of previous methods.

Filtered back projection

Initially, FBP was the standard for CT reconstruction, valued for its computational efficiency and accurate spatial resolution in high-dose environments [5]. Developed in the 1970s, FBP mathematically reconstructs images by first applying a convolution filter—such as a ramp or modified filter—to raw projection data (sinograms) to correct for blurring, and then back-projecting the filtered data across the image matrix [15]. This linear reconstruction approach enables extremely rapid image generation, making it ideal for urgent or high-throughput situations. However, the linear nature of FBP reconstruction makes it highly sensitive to statistical photon noise, which increases inversely with the square root of radiation dose [4,5]. As a result, low-dose scans often exhibit grainy images that obscure fine anatomical details. FBP is also prone to artifacts caused by patient motion, beam hardening from dense structures (e.g., bone, metallic implants), and incomplete projection data, all of which can degrade diagnostic quality. These limitations restrict the effectiveness of FBP in complex clinical scenarios, frequently necessitating higher radiation doses to maintain acceptable image quality. The need to overcome these challenges—particularly in dose-sensitive populations—drove the development of advanced reconstruction methods.

Iterative reconstruction

Developed in the 2000s, IR was introduced to address the noise and artifact limitations of FBP by modelling the imaging process iteratively [9,10,17,18]. Unlike FBP's single-step back-projection, IR employs a multi-step optimization loop. The process begins with an initial image estimate—often generated by FBP—which is forward-projected to simulate the raw projection data (sinogram) that the scanner would acquire. This simulated data is then compared to the actual measured projection data, and the difference (the "residual") is used to update the image estimate. This forward- and back-projection cycle is repeated until the difference between simulated and measured data falls below a predefined threshold or the maximum iteration count is reached. This process effectively improves noise suppression and reduces artifacts.

Adaptive Statistical Iterative Reconstruction (ASIR) was an early approach that incorporated statistical noise models to improve image quality while maintaining reconstruction efficiency [8]. ASIR applies a hybrid approach, blending FBP data with statistically modelled noise suppression, to speed convergence and reduce the "plastic" texture sometimes seen with full iterative methods. More advanced techniques, such as Model-Based Iterative Reconstruction (MBIR), further enhance accuracy by incorporating detailed physical and statistical models of the imaging system, including x-ray photon statistics, system optics, and detector response [19]. MBIR explicitly models the system's geometry, focal spot size, detector aperture, and scatter properties, allowing it to suppress noise more effectively and reduce beam hardening, streak, and partial-volume artifacts [20].

However, IR can give images a "plastic" appearance, which may reduce diagnostic confidence. Furthermore, the computational demands of MBIR can prolong reconstruction times, making it less practical in emergency situations [12]. These textural changes occur because aggressive noise suppression can obscure fine image granularity, an effect that, while improving quantitative noise metrics, may paradoxically lower radiologists' subjective confidence. This drawback, along with hardware demands, was a major impetus for the development of DLR, which aims to combine IR's noise reduction with the more natural image texture of FBP.

Deep learning reconstruction

DLR represents the latest stage in CT reconstruction evolution, designed to combine the low-noise performance of IR with the natural image texture and speed of FBP. Using deep neural networks (DNNs) trained on high-quality reference images, DLR distinguishes anatomical signal from noise and artifacts [21–23]. This enables reconstructions that maintain diagnostic detail even at substantially reduced radiation doses, a benefit particularly important for pediatric, oncology, and screening populations. Compared to IR, DLR preserves image granularity similar to high-dose FBP, which improves radiologist confidence [24], and can reconstruct full datasets in seconds, making it suitable for both routine workflows and time-critical scenarios such as stroke or trauma [25].

Technical foundations of DLR in CT

Integration approaches

DLR can be integrated at multiple stages of the CT image formation pipeline, each with specific advantages [26,27]:

- **Direct DLR:** It replaces traditional reconstruction methods like FBP or IR, using neural networks to generate images directly from raw projection data. By learning the relationship between projection data and high-quality images, this approach can compensate for missing or undersampled measurements (e.g., sparse-view acquisitions or

truncated scans), thereby reducing artifacts and noise that would otherwise arise from incomplete scan data [28].

- **Projection-space DLR:** Noise and artifacts such as beam hardening and scatter are reduced before conventional reconstruction, improving low-dose and high-attenuation imaging [29].
- **Image-space DLR:** Applied after FBP or IR reconstruction, deep learning refines the image to reduce noise and enhance clarity, typically using supervised training with paired high- and low-dose images [30].
- **Hybrid DLR:** Combines projection- and image-space methods to balance computational efficiency with quality, while retaining compatibility with existing reconstruction frameworks.

Deep neural network architecture and training

Most DLR systems employ multi-layer convolutional neural networks that first extract basic spatial features, then progressively refine anatomical detail while suppressing structured and random noise [31]. Supervised models are trained with paired high- and low-dose CT datasets, using high-dose images as “ground truth” to iteratively update model weights via backpropagation. Unsupervised or semi-supervised approaches (e.g., CycleGAN) learn structural consistency without paired datasets, improving adaptability when matched pairs are scarce [32,33]. Validation against multi-center, multi-vendor datasets is critical to ensure robustness in varied clinical conditions.

Adjustable strength levels and real-time integrations

Many commercially available DLR implementations (e.g., GE TrueFidelity™) offer three adjustable reconstruction strength settings—Low, Medium, and High—to meet varying clinical needs [26]. In such systems, the Low setting emphasizes detail preservation, making it suitable for high-detail tasks such as lung or bone imaging; the Medium setting provides a balance between noise suppression and texture preservation for general use; and the High setting maximizes noise reduction, which is often preferred for low-dose protocols such as pediatric imaging. These strength-level frameworks are vendor-specific and may vary in number, naming, and algorithmic behaviour across different manufacturers. Once integrated into CT consoles, DLR processes projection data in a single forward pass, generating DICOM images in seconds and supporting additional pipelines such as metal artifact reduction or dual-energy processing.

Clinical validation and adoption

Multi-center validation studies have confirmed DLR’s benefits across neuro, thoracic, abdominopelvic, cardiovascular, and musculoskeletal CT [34,35]. These include improved gray–white matter differentiation in acute stroke, higher conspicuity of pulmonary nodules and emboli, better stent lumen visualization, and superior lesion detection in low-contrast abdominal organs—all at reduced dose levels. This consistent performance across anatomical regions has accelerated adoption in both academic and community practice.

Clinical applications of DLR in CT

DLR technology holds transformative potential in numerous CT applications, improving diagnostic quality while reducing radiation exposure. Its proficiency in decreasing noise, minimizing artifacts, and preserving natural image textures is invaluable across diverse clinical scenarios. The following sections explore the applications of DLR in neuroimaging, chest CT, abdominal and pelvic CT, cardiovascular imaging, pediatric imaging, and low-dose screening.

Neuroimaging

DLR is highly beneficial in neuroimaging, where distinguishing subtle density differences, such as those between gray and white matter, is crucial [36]. It enhances gray-white matter differentiation, essential for evaluating ischemic changes, traumatic injuries, and neurodegenerative diseases [37–39]. For instance, during the assessment of an acute infarct, DLR delivers superior contrast between gray and white matter compared to traditional methods like FBP and IR, as illustrated in Fig. 1. Such improvement in tissue contrast may assist radiologists in visualizing subtle parenchymal changes. Additionally, DLR effectively reduces artifacts in challenging areas such as the skull base and posterior fossa, which are frequently affected by beam hardening or noise. Its application in these regions proves particularly valuable for evaluating traumatic brain injuries or posterior fossa lesions, thus improving the utility of low-dose neuroimaging in emergency and pediatric situations [40].

Chest imaging

DLR provides significant benefits in chest CT, an essential tool for evaluating a variety of thoracic conditions. By efficiently reducing noise and maintaining anatomical details, DLR enhances diagnostic accuracy for both parenchymal and vascular abnormalities [41,42]. In the context of pulmonary embolism (Fig. 2), studies have reported that DLR improves subjective image quality and vessel conspicuity in CT pulmonary angiography compared to FBP and IR, particularly in segmental and subsegmental branches, without compromising diagnostic confidence [43,44].

In patients with emphysema, DLR excels at visualizing subtle parenchymal changes and enhancing diagnostic confidence, as illustrated in Fig. 3 [45]. Similarly, in cases of pneumonia, DLR sharpens the delineation of consolidation boundaries and associated air bronchograms, even at reduced doses. In chronic obstructive pulmonary disease (COPD), patients often undergo multiple follow-up CT scans over time to monitor emphysematous changes, airway wall thickening, or complications such as superimposed infection. By enabling high-quality imaging at lower radiation doses, DLR can help reduce cumulative radiation exposure while preserving the visibility of subtle parenchymal and airway abnormalities. Furthermore, DLR’s ability to maintain high-quality imaging in low-dose protocols supports its use in lung cancer screening, where early detection of small pulmonary nodules is critical for improving patient outcomes [46].

Cardiovascular imaging

DLR offers significant benefits in cardiovascular imaging, which requires exceptional resolution and clarity to evaluate intricate cardiac and vascular structures [47,48]. In cases of coronary artery disease, DLR enhances the visualization of calcified plaques and improves the evaluation of stented coronary arteries, while minimizing the streak artifacts commonly associated with high-density materials (Fig. 4 and Fig. 5) [49–51]. This capability is crucial for assessing stenosis, plaque burden, and post-intervention complications with greater accuracy.

Additionally, DLR exhibits superior performance in delayed contrast-enhanced cardiac CT, enhancing the detection of myocardial fibrosis (Fig. 6) [52]. By preserving high image fidelity and minimizing noise, DLR facilitates precise delineation of fibrotic regions, which is crucial for diagnosing cardiomyopathies and guiding treatment decisions. DLR is also essential for imaging prosthetic heart valves and pacemaker leads [53]. Its effectiveness in reducing metal-related artifacts allows for clear visualization of prosthetic aortic valves and the calcifications around pacemaker leads (Fig. 7 and Fig. 8) and makes it valuable for preoperative planning and postoperative monitoring in patients with cardiac devices. DLR further enhances aortic and congenital heart disease imaging. In aortic dissection, it provides superior noise suppression while preserving vessel texture, improving true and false lumen differentiation

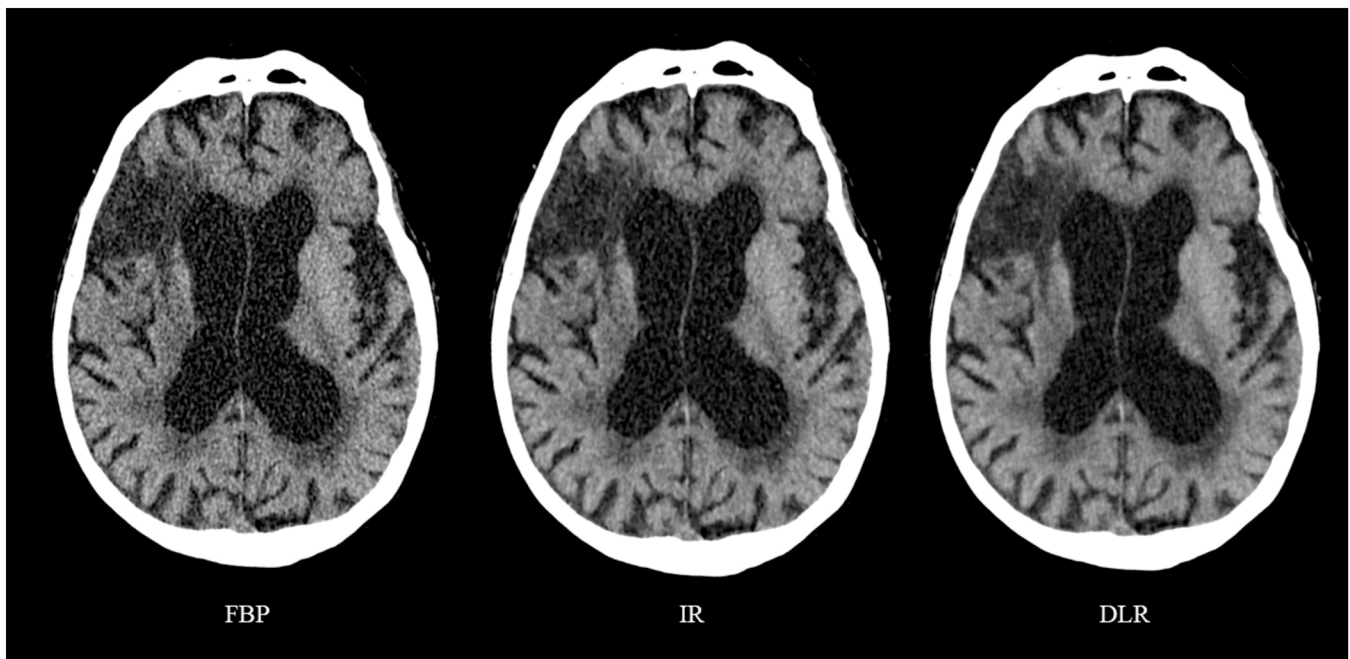


Fig. 1. Non-contrast CT head images reconstructed using FBP, IR, and DLR. DLR reconstruction enhances gray-white matter differentiation, improving the detection of subtle ischemic changes compared to FBP and IR.



Fig. 2. Contrast-enhanced CT chest images demonstrating a pulmonary embolism in the left lower lobe arteries (arrow) with an associated right pleural effusion. DLR reconstruction provides superior vascular detail and noise reduction compared to FBP and IR, enhancing diagnostic confidence in evaluating pulmonary embolism and related findings.

compared to FBP and IR (Fig. 9) [54]. In case of atrial septal defect imaging, DLR achieves optimal noise reduction while maintaining anatomical clarity, enhancing defect visualization in both four-chamber and oblique views (Fig. 10) [55]. These advantages underscore DLR's role in refining cardiovascular imaging, offering clearer, more diagnostically valuable images.

Abdominal and pelvic imaging

In abdominal and pelvic imaging, DLR enhances soft-tissue contrast, which is essential for detecting and characterizing lesions in organs such as the liver, pancreas, and kidneys [56–58]. It effectively reduces noise in low-contrast areas, improving the visibility of subtle lesions, such as hepatic metastases or cysts [59]. As demonstrated in Fig. 11, DLR provides superior noise reduction and lesion clarity in hepatic metastases

from colon cancer compared to FBP and IR, while preserving a more natural image texture. Similarly, in pancreatic imaging, DLR enhances visualization of locally advanced pancreatic adenocarcinoma and its vascular involvement, as shown in Fig. 12 [60]. This improvement is particularly valuable in oncology, where repeated follow-up scans are required for disease monitoring [61]. By decreasing cumulative radiation exposure without compromising diagnostic quality, DLR plays a crucial role in improving patient safety while maintaining high diagnostic accuracy.

Pediatric imaging

In pediatric imaging, minimizing radiation exposure is critical due to children's increased sensitivity to ionizing radiation [62,63]. DLR effectively meets this challenge by providing high-quality images at

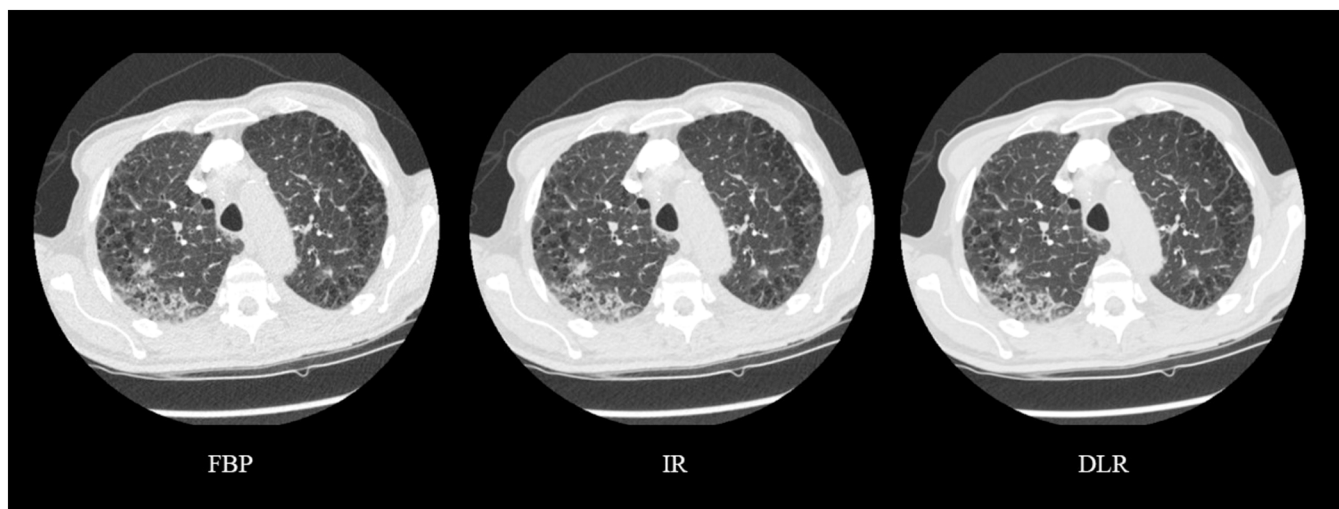


Fig. 3. High Resolution CT chest images showing emphysematous changes and pneumonic consolidations in the posterior segment of the right upper lobe. DLR reconstruction improves visualization of subtle parenchymal details and consolidations compared to FBP and IR, facilitating more accurate diagnosis.

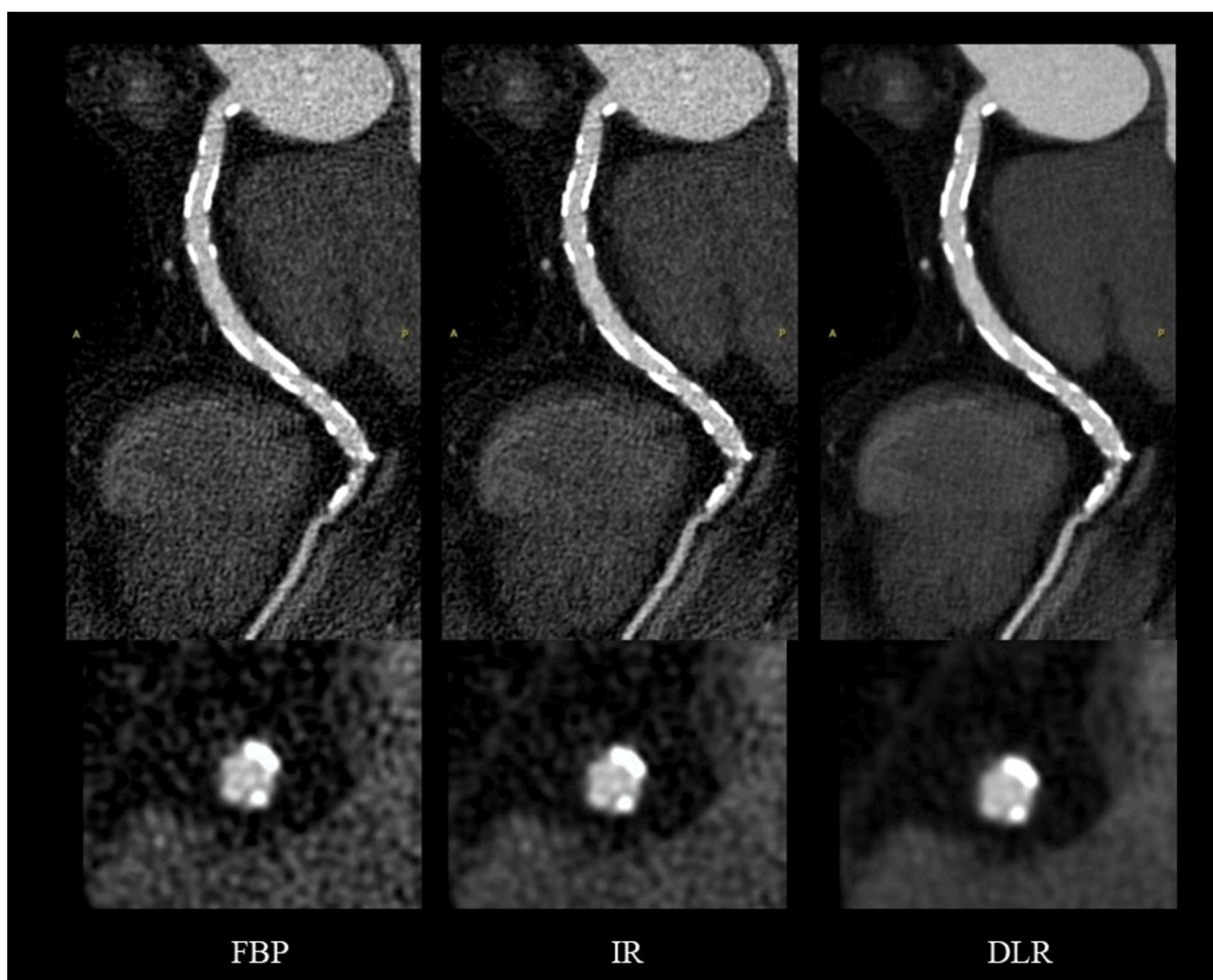


Fig. 4. CT coronary angiography images illustrating calcified coronary arteries reconstructed with FBP, IR, and DLR. DLR provides superior noise reduction while preserving vessel detail, improving the assessment of coronary artery calcifications. The enhanced clarity and artifact suppression with DLR allow for a more accurate evaluation of luminal narrowing and plaque burden.

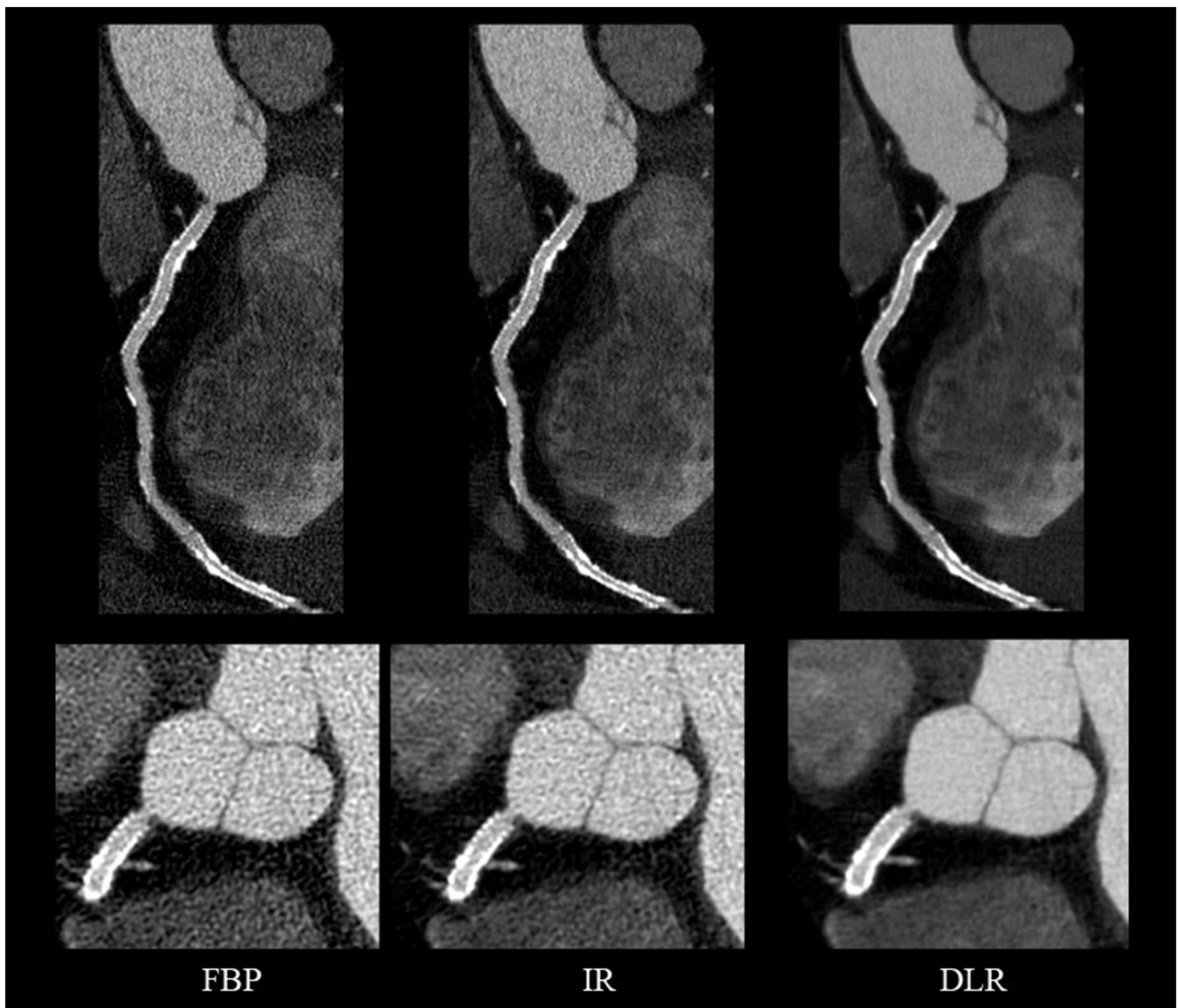


Fig. 5. CT coronary angiography images of coronary arteries with stents reconstructed using FBP, IR, and DLR. DLR reconstruction demonstrates superior artifact reduction and enhanced visualization of the stented lumen compared to FBP and IR, facilitating improved assessment of in-stent restenosis.

significantly lower doses, making it an indispensable tool for evaluating developmental and congenital abnormalities [64]. For instance, in pediatric neuroimaging, DLR allows for precise assessment of conditions such as hydrocephalus and intracranial tumors without subjecting patients to excessive radiation. Similarly, in pediatric cardiac CT, DLR enables detailed visualization of coronary arteries and congenital anomalies while ensuring safety and diagnostic reliability [65]. The dose optimization provided by DLR also benefits routine follow-up studies for conditions like neuroblastoma or congenital heart disease. By reducing cumulative radiation exposure, DLR aligns with safety standards, reassuring clinicians and families about the long-term implications of repeated imaging.

Low-dose screening and follow-up imaging

DLR's ability to produce high-quality images at low doses is increasingly applied in preventive and follow-up imaging [66]. In lung cancer screening, DLR supports low-dose protocols by ensuring clear visualization of nodules, contributing to early detection while minimizing radiation exposure. For osteoporosis and bone imaging, DLR

enhances bone clarity, aiding assessments in high-risk populations without increasing radiation levels [67]. In chronic disease monitoring—such as for inflammatory bowel disease or chronic liver disease—DLR reduces cumulative radiation from frequent follow-ups, thereby improving patient safety while maintaining diagnostic accuracy.

Performance comparison with other reconstruction methods

DLR offers significant advantages over traditional CT reconstruction methods, such as FBP and IR [11,68]. Evaluations across key metrics—noise reduction, Contrast-to-noise ratio, artifact suppression, image texture, and dose reduction—highlight DLR's superior performance in various clinical applications (Table 1).

Noise reduction and contrast-to-noise ratio

DLR provides considerable noise reduction compared to FBP, enhancing image clarity, particularly in low-contrast areas, while allowing for lower radiation doses [69]. Compared to IR methods, DLR

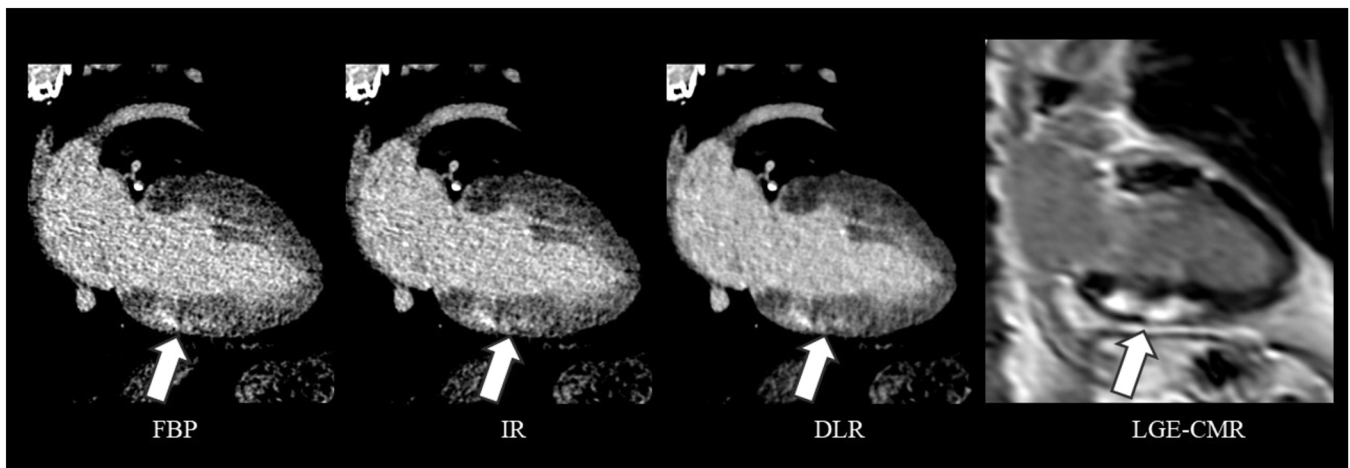


Fig. 6. Delayed contrast-enhanced cardiac CT images reconstructed using FBP, IR, and DLR. DLR provides superior noise reduction and an improved Contrast-to-noise ratio, enhancing the detection of myocardial fibrosis compared to FBP and IR. The improved tissue differentiation with DLR allows for better visualization of fibrotic regions. Corresponding area of myocardial fibrosis is seen as late gadolinium enhancement (LGE) in the basal myocardium on cardiac MR (CMR).

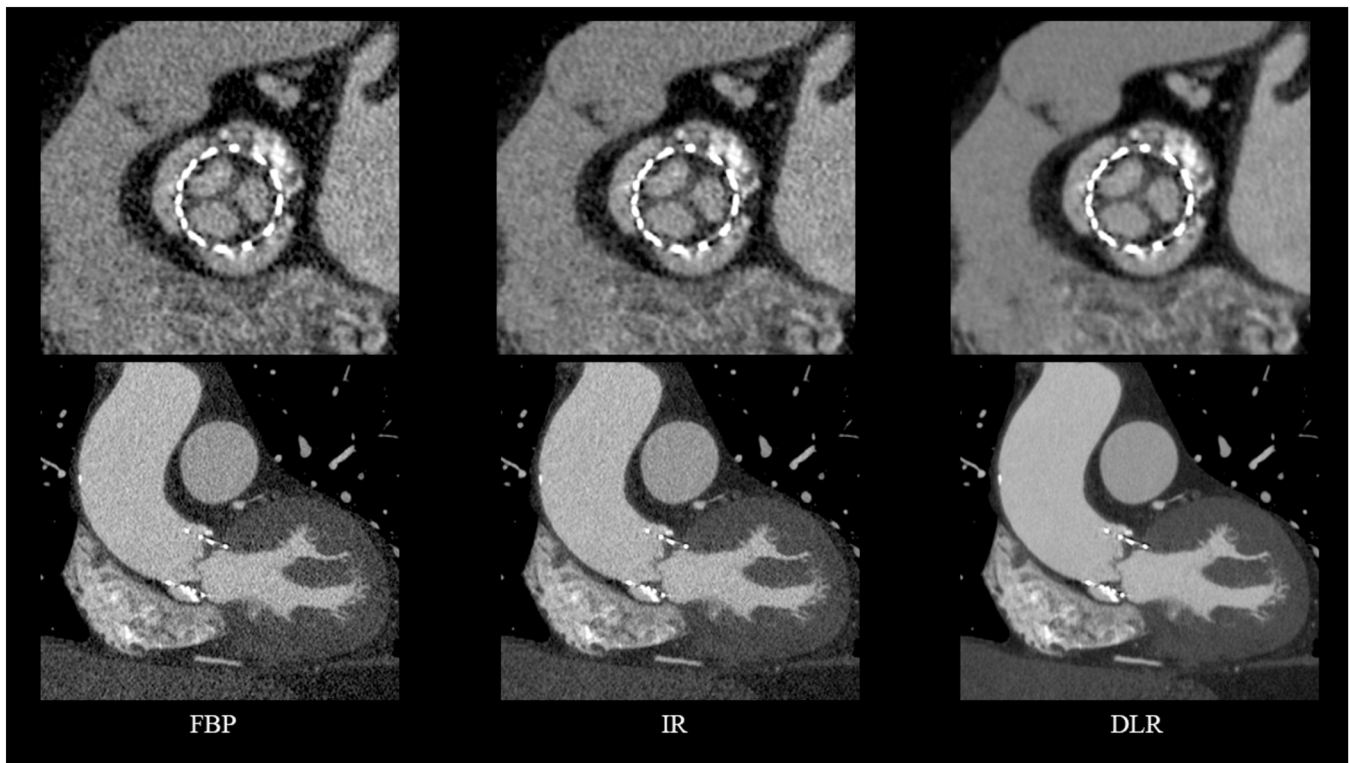


Fig. 7. CT cardiac angiography images of a prosthetic aortic valve post-TAVR, reconstructed using FBP, IR, and DLR. DLR provides improved visualization of the prosthetic valve, enabling a more accurate assessment of valve positioning, perivalvular leaks, and surrounding vascular structures compared to FBP and IR.

maintains natural image texture and achieves higher CNR without the artificial appearance often associated with iterative techniques. Quantitative comparisons from the literature demonstrate that DLR consistently reduces image noise and improves CNR across anatomical regions [70]. For example, Jensen et al. reported a 47 % reduction in noise and a 92–94 % increase in CNR for high-strength DLR compared to 30 % IR in abdominal CT [71]. In ultra-low-dose abdominal imaging, DLR achieved total noise levels (~ 7.6 HU) substantially lower than ultra-low-dose-hybrid IR (~ 12.1 HU) when measured at anterior abdominal wall subcutaneous fat, whereas the SNR were highest with the DLR compared to IR in various abdominal organs such as Liver (7.7 vs 4.1), Spleen (6.9 vs 3.8), Kidney (4.6 vs 2.5) [72]. In low-dose CT

colonography, DLR yielded the lowest noise (7.3 vs. 17.8) and highest SNR (4.2 vs. 1.6) and CNR (132.7 vs. 50.6) compared to IR [73]. CTA Brain also revealed similar results, for example the middle cerebral artery had lowest noise in DLR-M (7.9) and DLR-H (5.4) compared to IR (13.7) and FBP (17.3) [74].

Artifact suppression

Artifacts, such as beam hardening, motion artifacts, and streaks near high-density structures, pose common challenges in CT imaging [75]. DLR, leveraging deep neural networks, effectively distinguishes noise from anatomical details, outperforming IR in artifact suppression. This

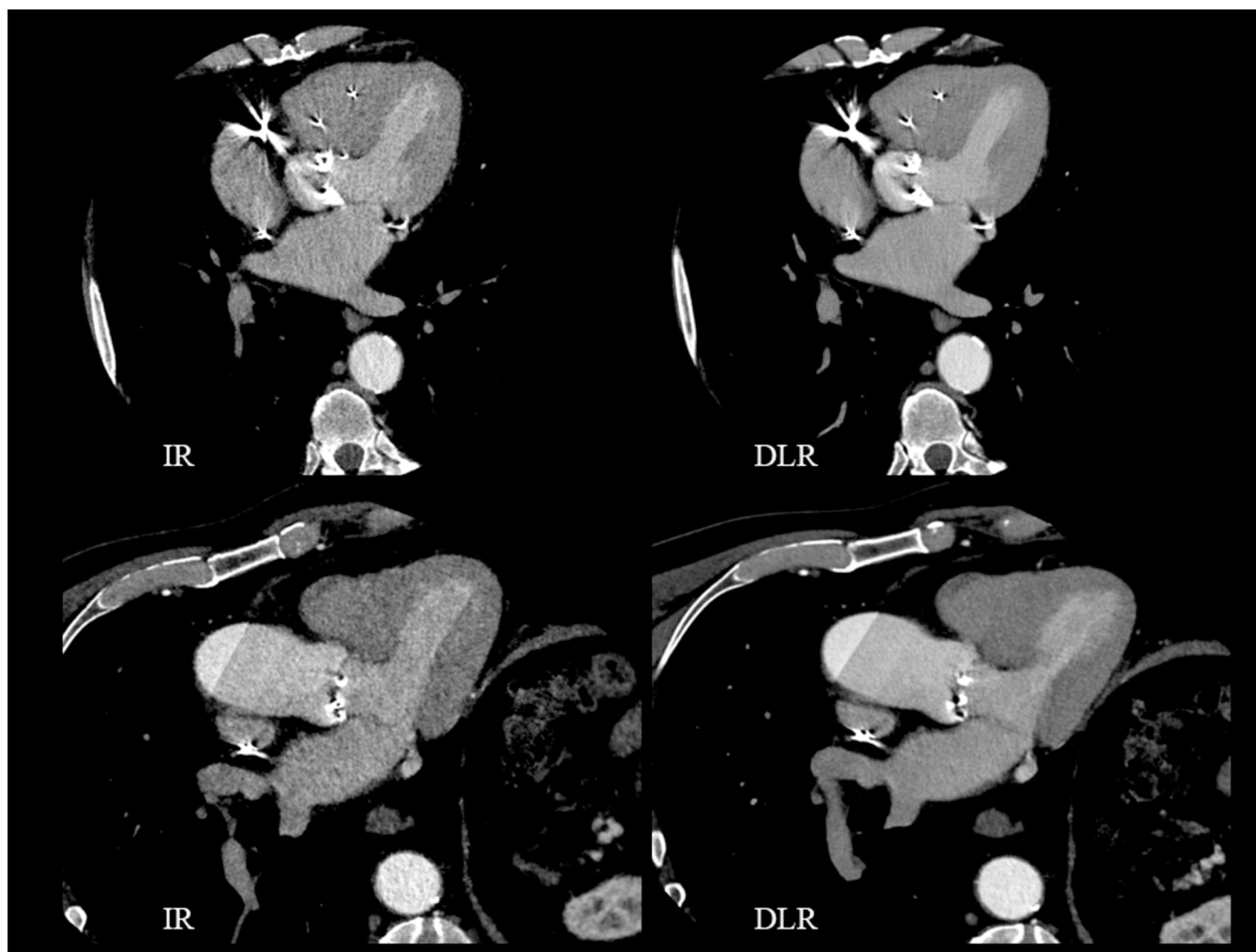


Fig. 8. Arterial-phase CT aortography images reconstructed using IR and DLR. DLR reconstruction demonstrates improved noise reduction and artifact suppression, enhancing visualization of aortic calcifications and surrounding vasculature while reducing metal-related artifacts from the pacemaker. These improvements facilitate a more accurate assessment of aortic pathology and vascular integrity.

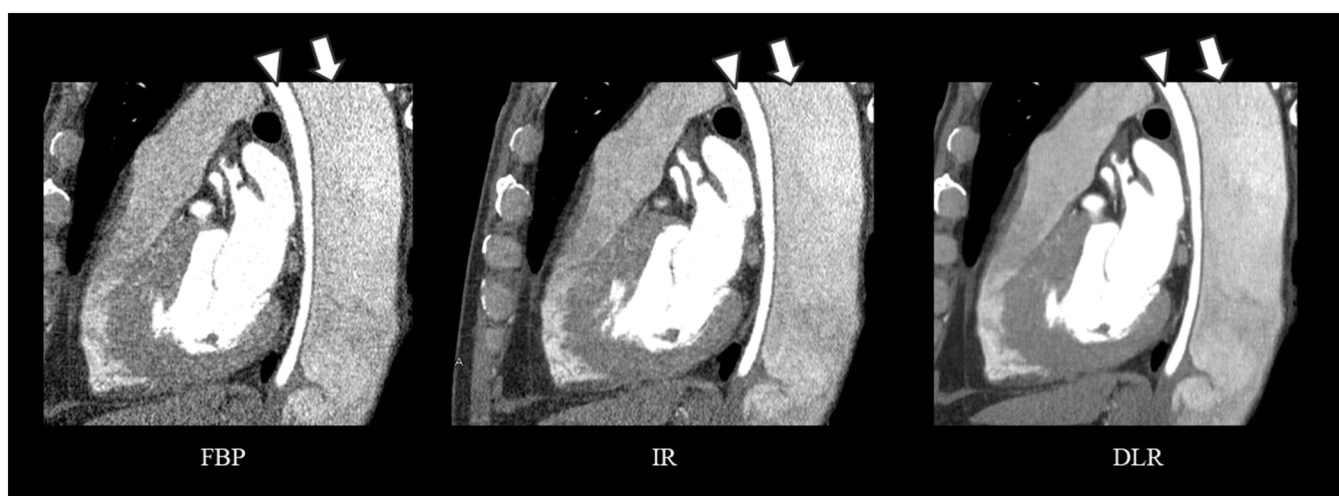


Fig. 9. Sagittal CT images of an aortic dissection reconstructed using FBP, IR, and DLR. The true lumen (arrowhead) and false lumen (arrow) are depicted. FBP exhibits high noise, IR reduces noise but smooths fine details, while DLR preserves texture and enhances lumen differentiation with superior noise suppression.



Fig. 10. Four-chamber and oblique-view cardiac CT angiography images demonstrating an atrial septal defect (ASD) using FBP, IR, and DLR reconstruction. DLR achieves optimal noise suppression while preserving anatomical definition, enhancing ASD visualization with improved contrast and natural image texture.

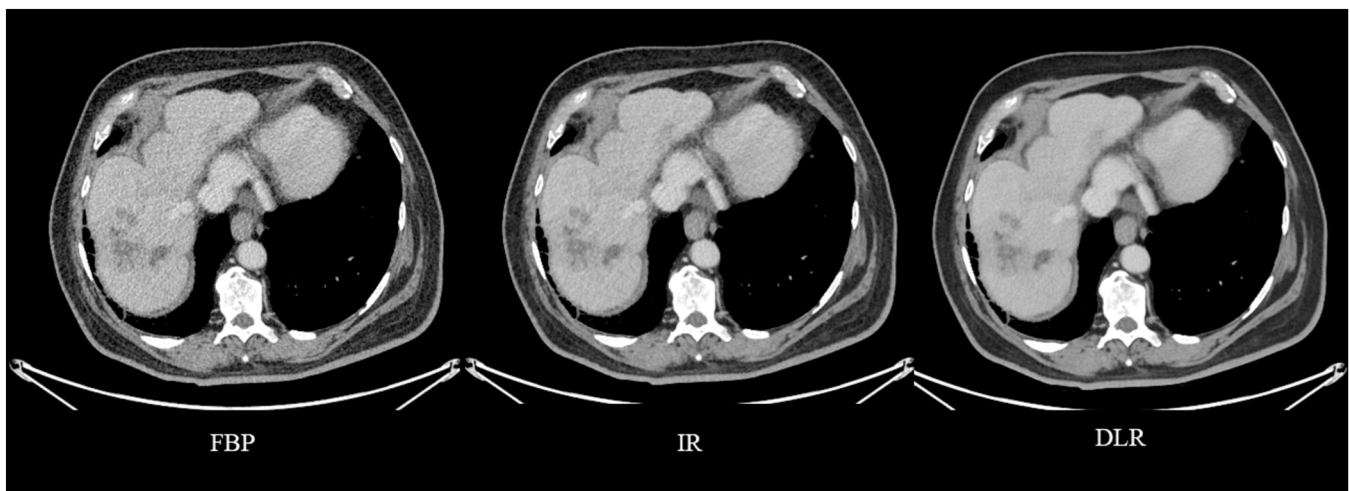


Fig. 11. Axial contrast-enhanced CT images of hepatic metastases from colon cancer reconstructed using FBP, IR, and DLR. DLR reconstruction provides superior noise reduction and lesion clarity while preserving natural texture, facilitating improved lesion detection.

capability is particularly beneficial in cardiovascular CT, where DLR lessens streak artifacts around calcified plaques and metal implants, enhancing the clarity of coronary arteries and other dense structures.

Image texture and natural appearance

A common drawback of IR is its “plastic” or artificial look, which can undermine radiologists’ confidence in interpreting images [10]. DLR is designed to preserve the natural texture of high-dose images, making them closely resemble those produced by FBP. This balance of realism and clarity enhances radiologist confidence, especially in neuro and

chest imaging where fine detail is critical.

Radiation dose reduction

DLR allows for high-quality imaging at lower radiation doses, making it ideal for dose-sensitive populations, including pediatric patients, those undergoing oncology treatment, and individuals in lung cancer screening programs [63,76]. Unlike FBP, which requires higher doses to maintain quality, DLR achieves comparable or superior image quality at significantly reduced doses. DLR avoids the texture degradation sometimes seen with IR at low doses, supporting safer routine and follow-up

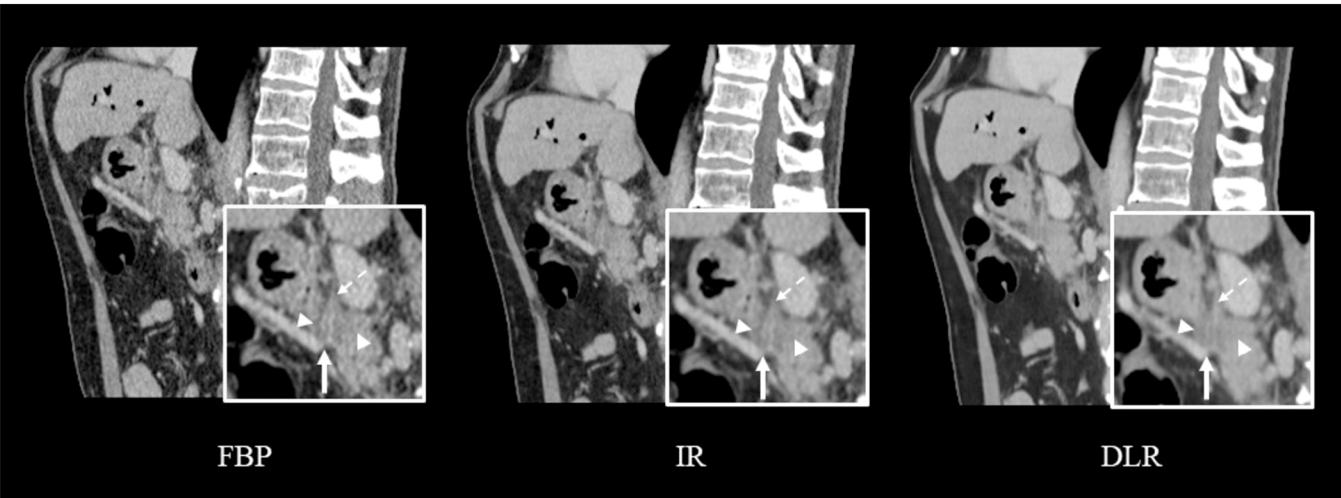


Fig. 12. Sagittal contrast-enhanced CT images of a locally advanced pancreatic adenocarcinoma (arrowheads) with tumor contact with the gastro-omental vein (arrow). Compared to FBP and IR, DLR reconstruction offers superior noise reduction while maintaining fine anatomical details and a more natural texture, enhancing visualization of the tumor and its vascular involvement. Notably, a small irregular duodenopancreatic vein running through the tumor is better visualized (dotted arrow).

Table 1
Comparative summary of key performance characteristics of CT reconstruction methods: FBP, IR, and DLR.

Feature	FBP	IR	DLR
Noise reduction	Poor	Moderate–Good	Excellent
Artifact suppression	Poor	Good	Excellent
Texture preservation	Good	Poor (“plastic”)	Good (natural)
Dose efficiency	Low	Moderate	High
Reconstruction time	Fast	Slow–Moderate	Fast (image-space DLR)

imaging.

Diagnostic confidence and clinical utility

The advantages of DLR—such as effective noise reduction, artifact suppression, texture preservation, and dose efficiency—collectively enhance diagnostic confidence [77,78]. Radiologists report increased confidence in detecting small lesions and vascular structures with DLR images. Additionally, DLR’s efficient integration into CT systems supports rapid clinical decision-making, which is vital in emergency situations. Its low-dose capability also expands its applicability across diverse patient populations, consistent with patient safety and quality care goals.

Current challenges and future directions in DLR

DLR has demonstrated clear advantages in CT imaging, but several practical and technical barriers still limit its routine use. A key issue is the heavy reliance on large, high-quality, and diverse training datasets [36,79,80]. Models developed on narrow or homogeneous data may not perform consistently in underrepresented populations or unusual anatomies, leading to reduced accuracy and potential bias. Multi-institutional collaborations, standardized protocols, and inclusion of a broad range of patient demographics can help improve generalizability [81,82]. Synthetic datasets generated from simulations could further address gaps for rare conditions and improve adaptability across different scanners and protocols [83].

Another important challenge is the “black-box” nature of DLR algorithms, which can make their decision-making process opaque to end users [84–86]. This lack of transparency may reduce clinical trust, especially when unfamiliar artifacts appear. Practical solutions include

integrating explainable AI tools that allow radiologists to view side-by-side pre- and post-DLR images, highlight the areas where the algorithm made the greatest changes, and flag low-confidence regions [87,88]. Embedding automated quality-control systems could also help maintain consistent performance across varying protocols.

DLR also demands substantial computing power for both training and deployment, often requiring high-performance GPUs or specialized processors [89]. These hardware needs can create cost barriers for smaller institutions and raise concerns about energy consumption and sustainability. Approaches such as model compression, cloud-based reconstruction, and more energy-efficient architectures could make DLR more accessible [90].

Although DLR effectively reduces most conventional CT artifacts, it can occasionally generate new, learned distortions or “hallucinated” features [91,92]. These are most often seen near metallic implants or dense calcifications, and may mislead interpretation if unrecognized [26]. To mitigate this, developers should train models on artifact-rich datasets and incorporate automated artifact-detection tools, while radiologists should be educated on identifying these novel patterns.

Finally, frequent algorithm updates and the need for large datasets present regulatory and ethical challenges [93,94]. Ongoing changes can complicate re-validation and regulatory approval, and the use of large imaging repositories raises questions about privacy, consent, and data security. Clear frameworks for continuous validation, transparent reporting of software updates, and robust ethical governance will be crucial.

Looking ahead, the role of DLR could expand beyond noise and artifact reduction. Integration with other AI-based tools for lesion detection, quantitative analysis, and personalized imaging protocols could streamline workflows and enhance diagnostic performance [95–97]. Adaptive, cross-platform models capable of adjusting to different scanners and acquisition settings [98] would further broaden applicability. Extending DLR principles to other modalities such as MRI and PET could bring consistent image quality improvements, shorter scan times, and better patient comfort across multi-modality imaging [99,100].

Conclusion

DLR represents a promising advancement in CT imaging, enabling high-quality reconstructions at reduced radiation doses while preserving natural image texture. By leveraging deep neural networks trained on

high-quality reference data, DLR offers improvements in noise suppression, artifact reduction, and image clarity compared to conventional methods such as FBP and IR. Although widespread adoption is currently limited by challenges in data diversity, interpretability and computational demands, ongoing technical refinement and collaborative validation efforts are steadily addressing these barriers. With continued progress, DLR has the potential to become an integral component of routine CT practice and, ultimately, a cornerstone technology in precision medical imaging.

Ethics statement

This manuscript is a review article and does not involve any studies with human participants or animals performed by any of the authors. No patient data or identifiable information has been included.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

CRediT authorship contribution statement

Asutosh Sahu: Writing – review & editing, Writing – original draft, Visualization, Conceptualization. **Corey T. Jensen:** Writing – review & editing, Supervision. **Ankush Jajodia:** Writing – review & editing. **Benoît Magnin:** Writing – review & editing, Visualization. **Benjamin Böttcher:** Writing – review & editing, Visualization. **Felix G. Meinel:** Writing – review & editing, Visualization. **Joji Ota:** Writing – review & editing, Visualization. **Hiroyuki Takaoka:** Writing – review & editing, Visualization. **Shobhit Mathur:** Writing – review & editing, Supervision.

Declaration of Competing Interest

Dr. Felix G. Meinel has received an institutional research grant and speaker's honoraria from GE Healthcare. Other co-authors have nothing to declare.

Acknowledgements

None.

Data availability

No data was used for the research described in the article.

References

- [1] J. Hsieh, T. Flohr, Computed tomography recent history and future perspectives, *J. Med. Imaging (Bellingham)* 8 (2021) 052109, <https://doi.org/10.1117/1.JMI.8.5.052109>.
- [2] J. Hsieh, B. Nett, Z. Yu, K. Sauer, J.-B. Thibault, C.A. Bouman, Recent advances in CT image reconstruction, *Curr. Radio. Rep.* 1 (2013) 39–51, <https://doi.org/10.1007/s40134-012-0003-7>.
- [3] M.J. Willemink, P.B. Noël, The evolution of image reconstruction for CT—from filtered back projection to artificial intelligence, *Eur. Radio.* 29 (2019) 2185–2195, <https://doi.org/10.1007/s00330-018-5810-7>.
- [4] R. Schofield, L. King, U. Tayal, I. Castellano, J. Stirrup, F. Pontana, J. Earls, E. Nicol, Image reconstruction: part 1 – understanding filtered back projection, noise and image acquisition, *J. Cardiovasc. Comput. Tomogr.* 14 (2020) 219–225, <https://doi.org/10.1016/j.jcct.2019.04.008>.
- [5] X. Pan, E.Y. Sidky, M. Vannier, Why do commercial CT scanners still employ traditional, filtered back-projection for image reconstruction? *Inverse Probl.* 25 (2009) 123009 <https://doi.org/10.1088/0266-5611/25/12/123009>.
- [6] W. Stiller, Basics of iterative reconstruction methods in computed tomography: a vendor-independent overview, *Eur. J. Radio.* 109 (2018) 147–154, <https://doi.org/10.1016/j.ejrad.2018.10.025>.
- [7] L.L. Geyer, U.J. Schoepf, F.G. Meinel, J.W. Nance, G. Bastarrika, J.A. Leipsic, N. S. Paul, M. Rengo, A. Laghi, C.N. De Cecco, State of the art: iterative CT reconstruction techniques, *Radiology* 276 (2015) 339–357, <https://doi.org/10.1148/radiol.2015132766>.
- [8] M.H. Goodenberger, N.A. Wagner-Bartak, S. Gupta, X. Liu, R.Q. Yap, J. Sun, E. P. Tamm, C.T. Jensen, Computed tomography image quality evaluation of a new iterative reconstruction algorithm in the abdomen (Adaptive Statistical Iterative Reconstruction–V) a comparison with Model-Based iterative reconstruction, adaptive statistical iterative reconstruction, and filtered back projection reconstructions, *J. Comput. Assist. Tomogr.* 42 (2018) 184–190, <https://doi.org/10.1097/RCT.0000000000000666>.
- [9] G. Li, X. Liu, C.T. Dodge, C.T. Jensen, X.J. Rong, A noise power spectrum study of a new model-based iterative reconstruction system: veo 3.0, *J. Appl. Clin. Med. Phys.* 17 (2016) 428–439, <https://doi.org/10.1120/jacmp.v17i5.6225>.
- [10] M.J. Willemink, P.A. de Jong, T. Leiner, L.M. de Heer, R.A.J. Nievelstein, R.P. J. Budde, A.M.R. Schilham, Iterative reconstruction techniques for computed tomography part 1: technical principles, *Eur. Radio.* 23 (2013) 1623–1631, <https://doi.org/10.1007/s00330-012-2765-y>.
- [11] M. Katsura, I. Matsuda, M. Akahane, J. Sato, H. Akai, K. Yasaka, A. Kunimatsu, K. Ohtomo, Model-based iterative reconstruction technique for radiation dose reduction in chest CT: comparison with the adaptive statistical iterative reconstruction technique, *Eur. Radio.* 22 (2012) 1613–1623, <https://doi.org/10.1007/s00330-012-2452-z>.
- [12] M.E. Telesmanich, C.T. Jensen, J.L. Enriquez, N.A. Wagner-Bartak, X. Liu, O. Le, W. Wei, A.G. Chandler, E.P. Tamm, Third version of vendor-specific model-based iterative reconstruction (Veo 3.0): evaluation of CT image quality in the abdomen using new noise reduction presets and varied slice optimization, *Br. J. Radio.* 90 (2017), <https://doi.org/10.1259/bjr.20170188>.
- [13] C.T. Dodge, E.P. Tamm, D.D. Cody, X. Liu, C.T. Jensen, W. Wei, V. Kundra, X. J. Rong, Performance evaluation of iterative reconstruction algorithms for achieving CT radiation dose reduction — a phantom study, *J. Appl. Clin. Med. Phys.* 17 (2016) 511–531, <https://doi.org/10.1120/jacmp.v17i2.5709>.
- [14] G. Wang, J.C. Ye, B. De Man, Deep learning for tomographic image reconstruction, *Nat. Mach. Intell.* 2 (2020) 737–748, <https://doi.org/10.1038/s42256-020-00273-z>.
- [15] R.A. Brooks, G. Di Chiro, Theory of image reconstruction in computed tomography, *Radiology* 117 (1975) 561–572, <https://doi.org/10.1148/117.3.561>.
- [16] P.J. La Riviere, C.R. Crawford, From EMI to AI: a brief history of commercial CT reconstruction algorithms, *J. Med. Imaging* 8 (2021), <https://doi.org/10.1117/1.JMI.8.5.052111>.
- [17] A. Padole, R.D. Ali Khawaja, M.K. Kalra, S. Singh, CT radiation dose and iterative reconstruction techniques, *Am. J. Roentgenol.* 204 (2015) W384–W392, <https://doi.org/10.2214/AJR.14.13241>.
- [18] C.T. Jensen, M.E. Telesmanich, N.A. Wagner-Bartak, X. Liu, J. Rong, J. Szklaruk, A. Qayyum, W. Wei, A.G. Chandler, E.P. Tamm, Evaluation of abdominal computed tomography image quality using a new version of Vendor-Specific Model-Based iterative reconstruction, *J. Comput. Assist. Tomogr.* 41 (2017) 67–74, <https://doi.org/10.1097/RCT.0000000000000472>.
- [19] L. Liu, Model-based iterative reconstruction: a promising algorithm for Today's computed tomography imaging, *J. Med. Imaging Radiat. Sci.* 45 (2014) 131–136, <https://doi.org/10.1016/j.jmir.2014.02.002>.
- [20] S. Boudabbous, D. Arditi, E. Paulin, A. Syrogiannopoulou, C. Becker, X. Montet, Model-Based iterative reconstruction (MBIR) for the reduction of metal artifacts on CT, *Am. J. Roentgenol.* 205 (2015) 380–385, <https://doi.org/10.2214/AJR.14.13334>.
- [21] H. Lan, Z. Du, DLIR: An Intermediate Representation for Deep Learning Processors, in: 2018: pp. 169–173. https://doi.org/10.1007/978-3-030-05677-3_19.
- [22] Y. Fu, Y. Lei, T. Wang, W.J. Curran, T. Liu, X. Yang, Deep learning in medical image registration: a review, *Phys. Med. Biol.* 65 (2020) 20TR01, <https://doi.org/10.1088/1361-6560/ab843e>.
- [23] M. Zhang, S. Gu, Y. Shi, The use of deep learning methods in low-dose computed tomography image reconstruction: a systematic review, *Complex Intell. Syst.* 8 (2022) 5545–5561, <https://doi.org/10.1007/s40747-022-00724-7>.
- [24] Y. Cheng, T.B. Smith, C.T. Jensen, X. Liu, E. Samei, Correlation of algorithmic and visual assessment of lesion detection in clinical images, *Acad. Radio.* 27 (2020) 847–855, <https://doi.org/10.1016/j.acra.2019.07.015>.
- [25] C. Arndt, F. Güttler, A. Heinrich, F. Bürckenmeyer, I. Diamantis, U. Teichgräber, Deep learning CT image reconstruction in clinical practice, *R. öFo Fortschr. Auf Dem Geb. Der R.öntgenstrahlen Und Der Bildgeb. Verfahr.* 193 (2021) 252–261, <https://doi.org/10.1055/a-1248-2556>.
- [26] J. Hsieh, E. Liu, B. Nett, J. Tang, J.-B. Thibault, S. Sahney, A new era of image reconstruction: TrueFidelity™ Technical white paper on deep learning image reconstruction, in: 2019. <https://api.semanticscholar.org/CorpusID:210167899>.
- [27] T. Liu, Z. He, DLIR: a deep learning-based initialization recommendation algorithm for trust-aware recommendation, *Appl. Intell.* 52 (2022) 11103–11114, <https://doi.org/10.1007/s10489-021-03039-1>.
- [28] Y. Li, K. Li, C. Zhang, J. Montoya, G.-H. Chen, Learning to reconstruct computed tomography images directly from sinogram data under a variety of data acquisition conditions, *IEEE Trans. Med. Imaging* 38 (2019) 2469–2481, <https://doi.org/10.1109/TMI.2019.2910760>.
- [29] L. Yang, Z. Li, R. Ge, J. Zhao, H. Si, D. Zhang, Low-Dose CT denoising via sinogram Inner-Structure transformer, *IEEE Trans. Med. Imaging* 42 (2023) 910–921, <https://doi.org/10.1109/TMI.2022.3219856>.
- [30] H. Chen, Y. Zhang, M.K. Kalra, F. Lin, Y. Chen, P. Liao, J. Zhou, G. Wang, Low-Dose CT with a residual Encoder-Decoder convolutional neural network, *IEEE*

- Trans. Med. Imaging 36 (2017) 2524–2535, <https://doi.org/10.1109/TMI.2017.2715284>.
- [31] P. Mohammadijrad, A. Mileto, L. Yu, S. Leng, L.S. Guimaraes, A.D. Missert, C. T. Jensen, H. Gong, C.H. McCollough, J.G. Fletcher, CT Noise-Reduction methods for Lower-Dose scanning: strengths and weaknesses of iterative reconstruction algorithms and new techniques, *RadioGraphics* 41 (2021) 1493–1508, <https://doi.org/10.1148/rq.2021200196>.
 - [32] X. Liang, L. Chen, D. Nguyen, Z. Zhou, X. Gu, M. Yang, J. Wang, S. Jiang, Generating synthesized computed tomography (CT) from cone-beam computed tomography (CBCT) using CycleGAN for adaptive radiation therapy, *Phys. Med. Biol.* 64 (2019) 125002, <https://doi.org/10.1088/1361-6560/ab22f9>.
 - [33] Z. Li, S. Zhou, J. Huang, L. Yu, M. Jin, Investigation of Low-Dose CT image denoising using unpaired deep learning methods, *IEEE Trans. Radiat. Plasma Med. Sci.* 5 (2021) 224–234, <https://doi.org/10.1109/TRPMS.2020.3007583>.
 - [34] J.H. Kim, H.J. Yoon, E. Lee, I. Kim, Y.K. Cha, S.H. Bak, Validation of Deep-Learning image reconstruction for Low-Dose chest computed tomography scan: emphasis on image quality and noise, *Korean J. Radio.* 22 (2021) 131, <https://doi.org/10.3348/kjr.2020.0116>.
 - [35] N. Lee, H.-H. Cho, S.M. Lee, S.K. You, Adaptation of deep learning image reconstruction for pediatric head CT: a focus on the image quality, *J. Korean Soc. Radiol.* 84 (2023) 240, <https://doi.org/10.3348/jksr.2021.0073>.
 - [36] M. Pula, E. Kucharczyk, A. Zdanowicz, M. Guzinski, Image quality improvement in deep learning image reconstruction of head computed tomography examination, *Tomography* 9 (2023) 1485–1493, <https://doi.org/10.3390/tomography9040118>.
 - [37] L. Lei, Y. Zhou, X. Guo, L. Wang, X. Zhao, H. Wang, J. Ma, S. Yue, The value of a deep learning image reconstruction algorithm in whole-brain computed tomography perfusion in patients with acute ischemic stroke, *Quant. Imaging Med. Surg.* 13 (2023) 8173–8189, <https://doi.org/10.21037/qims-23-547>.
 - [38] Z. Alagic, J. Diaz Cardenas, K. Halldorsson, V. Grozman, S. Wallgren, C. Suzuki, J. Helmenkamp, S.K. Koskinen, Deep learning versus iterative image reconstruction algorithm for head CT in trauma, *Emerg. Radio.* 29 (2022) 339–352, <https://doi.org/10.1007/s10140-021-02012-2>.
 - [39] I. Kim, H. Kang, H.J. Yoon, B.M. Chung, N.-Y. Shin, Deep learning-based image reconstruction for brain CT: improved image quality compared with adaptive statistical iterative reconstruction-Veo (ASIR-V), *Neuroradiology* 63 (2021) 905–912, <https://doi.org/10.1007/s00234-020-02574-x>.
 - [40] J. Sun, H. Li, B. Wang, J. Li, M. Li, Z. Zhou, Y. Peng, Application of a deep learning image reconstruction (DLIR) algorithm in head CT imaging for children to improve image quality and lesion detection, *BMC Med. Imaging* 21 (2021) 108, <https://doi.org/10.1186/s12880-021-00637-w>.
 - [41] Q. Tian, X. Li, J. Li, Y. Cheng, X. Niu, S. Zhu, W. Xu, J. Guo, Image quality improvement in low-dose chest CT with deep learning image reconstruction, *J. Appl. Clin. Med. Phys.* 23 (2022), <https://doi.org/10.1002/acm2.13796>.
 - [42] H. Wang, L.-L. Li, J. Shang, J. Song, B. Liu, Application of deep learning image reconstruction in low-dose chest CT scan, *Br. J. Radio.* 95 (2022), <https://doi.org/10.1259/bjr.20210380>.
 - [43] M. Lenfant, O. Chevallier, P.-O. Comby, G. Secco, K. Haioun, F. Ricolfi, B. Lemogne, R. Loffroy, Deep learning versus iterative reconstruction for CT pulmonary angiography in the emergency setting: improved image quality and reduced radiation dose, *Diagnostics* 10 (2020) 558, <https://doi.org/10.3390/diagnostics10080558>.
 - [44] A.-C. Klemenz, L. Albrecht, M. Mancke, A. Dalmer, B. Böttcher, A. Surov, M.-A. Weber, F.G. Meinel, Improved image quality in CT pulmonary angiography using deep learning-based image reconstruction, *Sci. Rep.* 14 (2024) 2494, <https://doi.org/10.1038/s41598-024-52517-2>.
 - [45] F. Ferri, R. Bouzerar, M. Auquier, J. Vial, C. Renard, Pulmonary emphysema quantification at low dose chest CT using deep learning image reconstruction, *Eur. J. Radio.* 152 (2022) 110338, <https://doi.org/10.1016/j.ejrad.2022.110338>.
 - [46] B. Jiang, N. Li, X. Shi, S. Zhang, J. Li, G.H. de Bock, R. Vliegthart, X. Xie, Deep learning reconstruction shows better lung nodule detection for Ultra-Low-Dose chest CT, *Radiology* 303 (2022) 202–212, <https://doi.org/10.1148/radiol.210551>.
 - [47] S.A. Koo, Y. Jung, K.A. Um, T.H. Kim, J.Y. Kim, C.H. Park, Clinical feasibility of deep Learning-Based image reconstruction on coronary computed tomography angiography, *J. Clin. Med.* 12 (2023) 3501, <https://doi.org/10.3390/jcm12103501>.
 - [48] C. Onnis, M. van Assen, E. Muscogiuri, G. Muscogiuri, G. Gershon, L. Saba, C. N. De Cecco, The role of artificial intelligence in cardiac imaging, *Radio. Clin. North Am.* 62 (2024) 473–488, <https://doi.org/10.1016/j.rcl.2024.01.002>.
 - [49] Y. Wang, G. Wang, X. Huang, W. Zhao, Q. Zeng, Y. Li, J. Guo, Improving image quality and resolution of coronary arteries in coronary computed tomography angiography by using high-definition scans and deep learning image reconstruction, *Quant. Imaging Med. Surg.* 13 (2023) 2933–2940, <https://doi.org/10.21037/qims-22-186>.
 - [50] A. Rossi, A.G. Gennari, D. Etter, D.C. Benz, T. Sartoretto, A.A. Giannopoulos, N. Mikail, S. Bengs, A. Maurer, C. Gebhard, R.R. Buechel, P.A. Kaufmann, T. A. Fuchs, M. Messerli, Impact of deep learning image reconstructions (DLIR) on coronary artery calcium quantification, *Eur. Radio.* 33 (2022) 3832–3838, <https://doi.org/10.1007/s00330-022-09287-0>.
 - [51] M. Liu, X. Chen, W. Liu, Y. Guo, Y. Zhu, Y. Duan, W. Huang, W. Kong, C. Yan, J. Qin, Effect of deep learning image reconstruction with high-definition standard scan mode on image quality of coronary stents and arteries, *Quant. Imaging Med. Surg.* 14 (2024) 1616–1635, <https://doi.org/10.21037/qims-23-1064>.
 - [52] S. Aoki, H. Takaoka, J. Ota, T. Kanaeda, T. Sakai, K. Matsumoto, Y. Noguchi, Y. Nishikawa, S. Yashima, K. Suzuki, K. Yoshida, M. Kinoshita, N. Suzuki-Eguchi, H. Sasaki, Y. Kobayashi, Strong diagnostic performance of single energy 256-row multidetector computed tomography with deep learning image reconstruction in the assessment of myocardial fibrosis, *Intern. Med.* 63 (2024) 2615–2623, <https://doi.org/10.2169/internalmedicine.2615-23>.
 - [53] H.-H. Cho, S.M. Lee, S.K. You, Assessment of deep learning image reconstruction (DLIR) on image quality in pediatric cardiac CT datasets type of manuscript: original research, *PLoS One* 19 (2024) e0300090, <https://doi.org/10.1371/journal.pone.0300090>.
 - [54] A. Heinrich, F. Streckenbach, E. Beller, J. Groß, M.-A. Weber, F.G. Meinel, Deep Learning-Based image reconstruction for CT angiography of the aorta, *Diagnostics* 11 (2021) 2037, <https://doi.org/10.3390/diagnostics11112037>.
 - [55] M. Gulizia, L. Alamo, Y. Alemán-Gómez, T. Cherpillod, K. Mandralis, C. Chevallier, E. Tenisch, A. Viry, Gated cardiac CT in infants: what can we expect from deep learning image reconstruction algorithm? *J. Cardiovasc. Comput. Tomogr.* 18 (2024) 304–306, <https://doi.org/10.1016/j.jcct.2024.03.001>.
 - [56] L. Cao, X. Liu, T. Qu, Y. Cheng, J. Li, Y. Li, L. Chen, X. Niu, Q. Tian, J. Guo, Improving spatial resolution and diagnostic confidence with thinner slice and deep learning image reconstruction in contrast-enhanced abdominal CT, *Eur. Radio.* 33 (2022) 1603–1611, <https://doi.org/10.1007/s00330-022-09146-y>.
 - [57] T. Kaga, Y. Noda, K. Fujimoto, T. Suto, N. Kawai, T. Miyoshi, F. Hyodo, M. Matsuo, Deep-learning-based image reconstruction in dynamic contrast-enhanced abdominal CT: image quality and lesion detection among reconstruction strength levels, *Clin. Radio.* 76 (2021) 710.e15–710.e24, <https://doi.org/10.1016/j.crad.2021.03.010>.
 - [58] T. Njølstad, A. Schulz, J.C. Godt, H.M. Brøgger, C.K. Johansen, H.K. Andersen, A. C.T. Martinsen, Improved image quality in abdominal computed tomography reconstructed with a novel deep learning image reconstruction technique – initial clinical experience, *Acta Radio. Open* 10 (2021), <https://doi.org/10.1177/20584601211008391>.
 - [59] C.T. Jensen, S. Gupta, M.M. Saleh, X. Liu, V.K. Wong, U. Salem, W. Qiao, E. Samei, N.A. Wagner-Bartak, Reduced-Dose deep learning reconstruction for abdominal CT of liver metastases, *Radiology* 303 (2022) 90–98, <https://doi.org/10.1148/radiol.211838>.
 - [60] P. Lyu, B. Neely, J. Solomon, F. Rigiroli, Y. Ding, F.R. Schwartz, B. Thomsen, C. Lowry, E. Samei, D. Marin, Effect of deep learning image reconstruction in the prediction of resectability of pancreatic cancer: diagnostic performance and reader confidence, *Eur. J. Radio.* 141 (2021) 109825, <https://doi.org/10.1016/j.ejrad.2021.109825>.
 - [61] P. Hou, N. Liu, X. Feng, Y. Chen, H. Wang, X. Wang, J. Liu, P. Zhan, X. Liu, B. Shang, Z. Shen, L. Wang, J. Gao, P. Lyu, A paradigm shift in oncology imaging: a prospective cross-sectional study to assess low-dose deep learning image reconstruction versus standard-dose iterative reconstruction for comprehensive lesion detection in dual-energy computed tomography, *Quant. Imaging Med. Surg.* 14 (2024) 6449–6465, <https://doi.org/10.21037/qims-24-197>.
 - [62] K. Zhang, X. Shi, S.-S. Xie, J.-H. Sun, Z.-H. Liu, S. Zhang, J.-Y. Song, W. Shen, Deep learning image reconstruction in pediatric abdominal and chest computed tomography: a comparison of image quality and radiation dose, *Quant. Imaging Med. Surg.* 12 (2022) 3238–3250, <https://doi.org/10.21037/qims-21-936>.
 - [63] C.K.C. Ng, Artificial intelligence for radiation dose optimization in pediatric radiology: a systematic review, *Children* 9 (2022) 1044, <https://doi.org/10.3390/children9071044>.
 - [64] J. Sun, H. Li, J. Li, T. Yu, M. Li, Z. Zhou, Y. Peng, Improving the image quality of pediatric chest CT angiography with low radiation dose and contrast volume using deep learning image reconstruction, *Quant. Imaging Med. Surg.* 11 (2021) 3051–3058, <https://doi.org/10.21037/qims-20-1158>.
 - [65] H.-H. Cho, S.M. Lee, S.K. You, Assessment of deep learning image reconstruction (DLIR) on image quality in pediatric cardiac CT datasets type of manuscript: original research, *PLoS One* 19 (2024) e0300090, <https://doi.org/10.1371/journal.pone.0300090>.
 - [66] C. Pozzessere, C. von Garnier, C. Beigelman-Aubry, Radiation exposure to Low-Dose computed tomography for lung cancer screening: should we be concerned? *Tomography* 9 (2023) 166–177, <https://doi.org/10.3390/tomography9010015>.
 - [67] Y. Li, Y. Jiang, X. Yu, B. Ren, C. Wang, S. Chen, D. Ma, D. Su, H. Tian, X. Ren, X. Yang, J. Gao, Y. Wu, Deep-learning image reconstruction for image quality evaluation and accurate bone mineral density measurement on quantitative CT: a phantom-patient study, *Front. Endocrinol. (Lausanne)* 13 (2022), <https://doi.org/10.3389/fendo.2022.884306>.
 - [68] O. Chandran, M. S. Pendem, P. P. S. C. Chacko, P.-R. Kadavigere, Influence of deep learning image reconstruction algorithm for reducing radiation dose and image noise compared to iterative reconstruction and filtered back projection for head and chest computed tomography examinations: a systematic review, *F1000Res* 13 (2024) 274, <https://doi.org/10.12688/f1000research.147345.1>.
 - [69] J. Zhong, Y. Xia, Y. Chen, J. Li, W. Lu, X. Shi, J. Feng, F. Yan, W. Yao, H. Zhang, Deep learning image reconstruction algorithm reduces image noise while alters radiomics features in dual-energy CT in comparison with conventional iterative reconstruction algorithms: a phantom study, *Eur. Radio.* 33 (2022) 812–824, <https://doi.org/10.1007/s00330-022-09119-1>.
 - [70] J.A. van Stiphout, J. Driessen, L.R. Koetzier, L.B. Ruules, M.J. Willemink, J.W. T. Heemskerk, A.J. van der Molen, The effect of deep learning reconstruction on abdominal CT densitometry and image quality: a systematic review and meta-analysis, *Eur. Radio.* 32 (2022) 2921–2929, <https://doi.org/10.1007/s00330-021-08438-z>.
 - [71] C.T. Jensen, X. Liu, E.P. Tamm, A.G. Chandler, J. Sun, A.C. Morani, S. Javadi, N. A. Wagner-Bartak, Image quality assessment of abdominal CT by use of new deep learning image reconstruction: initial experience, *Am. J. Roentgenol.* 215 (2020) 50–57, <https://doi.org/10.2214/AJR.19.22332>.

- [72] X. Zhang, G. Zhang, L. Xu, X. Bai, J. Zhang, M. Xu, J. Yan, D. Zhang, Z. Jin, H. Sun, Application of deep learning reconstruction of ultra-low-dose abdominal CT in the diagnosis of renal calculi, *Insights Imaging* 13 (2022) 163, <https://doi.org/10.1186/s13244-022-01300-w>.
- [73] Y. Chen, Z. Huang, L. Feng, W. Zou, D. Kong, D. Zhu, G. Dai, W. Zhao, Y. Zhang, M. Luo, Deep Learning-Based reconstruction improves the image quality of Low-Dose CT colonography, *Acad. Radio.* 31 (2024) 3191–3199, <https://doi.org/10.1016/j.acra.2024.01.021>.
- [74] X. Huang, J. Shang, Y. Xiao, W. Hou, G. Mu, D. Li, H. Qian, J. Li, Deep learning-based reconstruction improves image quality in low-dose head CT angiography, *Malawi Med. J.* 37 (2025) 109–114, <https://doi.org/10.4314/mmj.v37i2.8>.
- [75] J.F. Barrett, N. Keat, Artifacts in CT: recognition and avoidance, *RadioGraphics* 24 (2004) 1679–1691, <https://doi.org/10.1148/rg.246045065>.
- [76] E. Cui, Reducing radiation dose in routine CT scans: an AI-driven approach with deep learning-based dual-energy CT reconstruction, *Eur. Radio.* 34 (2023) 26–27, <https://doi.org/10.1007/s00330-023-10066-8>.
- [77] X. Lin, Y. Gao, C. Zhu, J. Song, L. Liu, J. Li, X. Wu, Improving diagnostic confidence in low-dose dual-energy CTE with low energy level and deep learning reconstruction, *Eur. J. Radio.* 178 (2024) 111607, <https://doi.org/10.1016/j.ejrad.2024.111607>.
- [78] Y. Bie, S. Yang, X. Li, K. Zhao, C. Zhang, H. Zhong, Impact of deep learning-based image reconstruction on image quality and lesion visibility in renal computed tomography at different doses, *Quant. Imaging Med. Surg.* 13 (2023) 2197–2207, <https://doi.org/10.21037/qims-22-852>.
- [79] J. Zhong, H. Shen, Y. Chen, Y. Xia, X. Shi, W. Lu, J. Li, Y. Xing, Y. Hu, X. Ge, D. Ding, Z. Jiang, W. Yao, Evaluation of image quality and detectability of deep learning image reconstruction (DLIR) algorithm in Single- and Dual-energy CT, *J. Digit. Imaging* 36 (2023) 1390–1407, <https://doi.org/10.1007/s10278-023-00806-z>.
- [80] S. Tripathi, K. Gabriel, S. Dheer, A. Parajuli, A.I. Augustin, A. Elahi, O. Awan, F. Dako, Understanding biases and disparities in radiology AI datasets: a review, *J. Am. Coll. Radiol.* 20 (2023) 836–841, <https://doi.org/10.1016/j.jacr.2023.06.015>.
- [81] P.M. Cheng, E. Montagnon, R. Yamashita, I. Pan, A. Cadrin-Chênevert, F. Perdigón Romero, G. Chartrand, S. Kadoury, A. Tang, Deep learning: an update for radiologists, *RadioGraphics* 41 (2021) 1427–1445, <https://doi.org/10.1148/rg.2021200210>.
- [82] G.-S. Hong, M. Jang, S. Kyung, K. Cho, J. Jeong, G.-Y. Lee, K. Shin, K.D. Kim, S. M. Ryu, J.B. Seo, S.M. Lee, N. Kim, Overcoming the challenges in the development and implementation of artificial intelligence in radiology: a comprehensive review of solutions beyond supervised learning, *Korean J. Radio.* 24 (2023) 1061, <https://doi.org/10.3348/kjr.2023.0393>.
- [83] T.P. Szczytkutowicz, B. Nett, L. Cherkezyan, M. Pozniak, J. Tang, M.G. Lubner, J. Hsieh, Protocol optimization considerations for implementing deep learning CT reconstruction, *Am. J. Roentgenol.* 216 (2021) 1668–1677, <https://doi.org/10.2214/AJR.20.23397>.
- [84] J. Zhong, L. Wang, C. Yan, Y. Xing, Y. Hu, D. Ding, X. Ge, J. Li, W. Lu, X. Shi, F. Yuan, W. Yao, H. Zhang, Deep learning image reconstruction generates thinner slice iodine maps with improved image quality to increase diagnostic acceptance and lesion conspicuity: a prospective study on abdominal dual-energy CT, *BMC Med Imaging* 24 (2024) 159, <https://doi.org/10.1186/s12880-024-01334-0>.
- [85] R. Jena, D. Sethi, P. Chaudhari, J.C. Gee, Deep Learning in Medical Image Registration: Magic or Mirage?, *ArXiv abs/2408.05839*, 2024. (<https://api.semanticscholar.org/CorpusID:271855238>).
- [86] A. Marey, P. Arjmand, A.D.S. Alerab, M.J. Eslami, A.M. Saad, N. Sanchez, M. Umair, Explainability, transparency and black box challenges of AI in radiology: impact on patient care in cardiovascular radiology, Egypt. J. Radiol. Nucl. Med. 55 (2024) 183, <https://doi.org/10.1186/s43055-024-01356-2>.
- [87] S. Cui, A. Traverso, D. Niraula, J. Zou, Y. Luo, D. Owen, I. El Naqa, L. Wei, Interpretable artificial intelligence in radiology and radiation oncology, *Br. J. Radio.* 96 (2023), <https://doi.org/10.1259/bjr.20230142>.
- [88] B.M. de Vries, G.J.C. Zwezerijnen, G.L. Burchell, F.H.P. van Velden, C.W. Menke-van der Houven van Oordt, R. Boellaard, Explainable artificial intelligence (XAI) in radiology and nuclear Medicine: a literature review, *Front. Med. (Lausanne)* 10 (2023), <https://doi.org/10.3389/fmed.2023.1180773>.
- [89] N.C. Thompson, K.H. Greenewald, K. Lee, G.F. Manso, The Computational Limits of Deep Learning, *CoRR abs/2007.05558*, 2020. (<https://arxiv.org/abs/2007.05558>).
- [90] D. Alis, T. Tanyel, E. Meltem, M.E. Seker, D. Seker, H.M. Karakas, E. Karaarslan, I. Öksüz, Choosing the right artificial intelligence solutions for your radiology department: key factors to consider, 0–0, *Diagn. Interv. Radiol.* 0 (2024), <https://doi.org/10.4274/dir.2024.232658>.
- [91] Y. Jung, Y.S. Song, I.S. Lee, S.J. Rhee, Comprehensive evaluation of artifact reduction and tissue recovery effects of metal artifact reduction technique based on full-reference metric, *Sci. Rep.* 13 (2023) 11679, <https://doi.org/10.1038/s41598-023-38516-9>.
- [92] J.J. Xu, L. Lönn, E. Budtz-Jørgensen, K.L. Hansen, P.S. Ulriksen, Quantitative and qualitative assessments of deep learning image reconstruction in low-keV virtual monoenergetic dual-energy CT, *Eur. Radio.* (2022), <https://doi.org/10.1007/s00330-022-09018-5>.
- [93] B. Murdoch, Privacy and artificial intelligence: challenges for protecting health information in a new era, *BMC Med. Ethics* 22 (2021) 122, <https://doi.org/10.1186/s12910-021-00687-3>.
- [94] F. Pesapane, C. Volonté, M. Codari, F. Sardanelli, Artificial intelligence as a medical device in radiology: ethical and regulatory issues in Europe and the United States, *Insights Imaging* 9 (2018) 745–753, <https://doi.org/10.1007/s13244-018-0645-y>.
- [95] Y. Wang, H. Zhan, J. Hou, X. Ma, W. Wu, J. Liu, J. Gao, Y. Guo, Y. Zhang, Influence of deep learning image reconstruction and adaptive statistical iterative reconstruction-V on coronary artery calcium quantification, –1726, *Ann. Transl. Med.* 9 (2021) 1726, <https://doi.org/10.21037/atm-21-5548>.
- [96] N. Nasrullah, J. Sang, M.S. Alam, M. Mateen, B. Cai, H. Hu, Automated lung nodule detection and classification using deep learning combined with multiple strategies, *Sensors* 19 (2019) 3722, <https://doi.org/10.3390/s19173722>.
- [97] J. Solomon, K. Dauber-Decker, S. Richardson, S. Levy, S. Khan, B. Coleman, R. Persaud, J. Chelico, D. King, A. Spyropoulos, T. McGinn, Integrating clinical decision support into electronic health record systems using a novel platform (EvidencePoint): developmental study, *JMIR Form. Res.* 7 (2023) e44065, <https://doi.org/10.2196/44065>.
- [98] V. Kulkarni, M. Gawali, A. Kharat, Key technology considerations in developing and deploying machine learning models in clinical radiology practice, *JMIR Med. Inf.* 9 (2021) e28776, <https://doi.org/10.2196/28776>.
- [99] G. Foti, C. Longo, Deep learning and AI in reducing magnetic resonance imaging scanning time: advantages and pitfalls in clinical practice, *Pol. J. Radio.* 89 (2024) 443–451, <https://doi.org/10.5114/pjr.192822>.
- [100] A. Artesani, A. Bruno, F. Gelardi, A. Chiti, Empowering PET: harnessing deep learning for improved clinical insight, *Eur. Radio. Exp.* 8 (2024) 17, <https://doi.org/10.1186/s41747-023-00413-1>.